

1. Soybean meal is 18% protein; cornmeal is 9% protein. How many pounds of each should be mixed together in order to get 360-lb mixture that is 17% protein?

How many pounds of cornmeal should be in the mixture?

pounds

How many pounds of soybean meal should be in the mixture?

pounds

2. A glass of skim milk supplies 0.1 mg of iron and 8.2 g of protein. A quarter pound of lean meat provides 3.4 mg of iron and 22 g of protein. A person on a special diet is to have 7.2 mg of iron and 76.8 g of protein. How many glasses of skim milk and how many quarter-pound servings of meat will provide this?

How many glass(es) of skim milk and how many quarter-pound serving(s) of lean meat would be required?

glass(es) of skim milk and quarter-pound serving(s) of lean meat will provide the needed nutrients.

- *3. An investment of \$37,000 was made by a business club. The investment was split into three parts and lasted for one year. The first part of the investment earned 8% interest, the second 6%, and the third 9%. Total interest from the investments was \$3030. The interest from the first investment was 6 times the interest from the second. Find the amounts of the three parts of the investment.

What is the amount of the first part of the investment? \$

What is the amount of the second part of the investment? \$

What is the amount of the third part of the investment? \$

4. Jake has \$200,000 to invest. He chooses one money market fund that pays 6.3% and a mutual fund that has more risk but has averaged 8.8% per year. If his goal is to average 6.8% per year with minimal risk, how much should he invest in each fund?

Jake should invest \$ in the money market fund that pays 6.3%.
(Simplify your answer.)

Jake should invest \$ in the mutual fund that averaged 8.8% per year.
(Simplify your answer.)

5. A theater owner wants to divide a 3900 seat theater into three sections, with tickets costing \$60, \$70, and \$90, depending on the section. He wants to have twice as many \$60 tickets as the sum of the other tickets, and he wants to earn \$259,000 from a full house. Find how many seats he should have in each section.

The number of \$60 seats is .

The number of \$70 seats is .

The number of \$90 seats is .

6. Ace Trucking Company has an order for delivery of three products, A, B, and C. The table gives the volume in cubic feet, the weight in pounds, and the value for insurance in dollars for a unit of each of the products. If the carrier can carry 12,600 cubic feet and 16,100 pounds and is insured for \$67,500, how many units of each product can be carried?

	Product A	Product B	Product C
Volume (cu ft)	24	20	80
Weight (lb)	40	30	60
Value (\$)	150	170	200

Select the correct choice below and fill in any answer boxes within your choice.

- ☐ A. There is one solution. The company can carry units of Product A, units of Product B, and units of Product C.
- ☐ B. There are infinitely many solutions. If the company carries z units of Product C, then the company can carry units of Product A and units of Product B.
- ☐ C. There is no solution.
7. The nutritional content per ounce of three foods is presented in the table below. If a meal consisting of the three foods allows exactly 3400 calories, 92 grams of protein, and 7750 milligrams of vitamin C, how many ounces of each kind of food should be used?

	Calories	Protein (in grams)	Vitamin C (in milligrams)
Food A	100	7	300
Food B	400	8	1250
Food C	500	12	150

Food A: oz

Food B: oz

Food C: oz